

PROBLEM SHEET 2: SOLUTIONS

1. (a) We have that a and c are the birth rates of x and y ; b and d are interaction parameters, effectively saying how good/bad interaction is for x and y ; η_1 and η_2 are the carrying capacities of x and y in the absence of the other, when both species exhibit logistic growth.
 - (b) Interactions are bad for x and good for y , so x is the prey and y is the predator in this model.
 - (c) The predator here can live without the prey, and in fact exhibits logistic growth up the carrying capacity. This suggests the predator has a more varied diet (these are sometimes called ‘generalist’ or ‘intraguild’ predators).
2. Setting $x = \hat{x}X$, $y = \hat{y}Y$, $t = \hat{t}T$, the system becomes

$$\frac{X}{T} \frac{d\hat{x}}{d\hat{t}} = a\hat{x}X \left(1 - \frac{\hat{x}X}{\eta_1}\right) - b\hat{x}\hat{y}XY, \quad (1a)$$

$$\frac{Y}{T} \frac{d\hat{y}}{d\hat{t}} = c\hat{y}Y \left(1 - \frac{\hat{y}Y}{\eta_2}\right) + d\hat{x}\hat{y}XY, \quad (1b)$$

Clearing the left-hand side of X , Y and T by multiplying through we get

$$\frac{d\hat{x}}{d\hat{t}} = a\hat{x}T \left(1 - \frac{\hat{x}X}{\eta_1}\right) - b\hat{x}\hat{y}TY, \quad (2a)$$

$$\frac{d\hat{y}}{d\hat{t}} = c\hat{y}T \left(1 - \frac{\hat{y}Y}{\eta_2}\right) + d\hat{x}\hat{y}TX. \quad (2b)$$

Keeping an eye on the target, we can bring the last term on the right-hand side for each equation into the bracket,

$$\frac{d\hat{x}}{d\hat{t}} = a\hat{x}T \left(1 - \frac{\hat{x}X}{\eta_1} - \frac{b}{a}\hat{y}Y\right), \quad (3a)$$

$$\frac{d\hat{y}}{d\hat{t}} = c\hat{y}T \left(1 - \frac{\hat{y}Y}{\eta_2} + \frac{d}{c}\hat{x}X\right), \quad (3b)$$

and now we can begin to compare term-by-term. Starting on the top, we can see $T = 1/a$ and $X = \eta_1$:

$$\frac{d\hat{x}}{d\hat{t}} = \hat{x} \left(1 - \hat{x} - \frac{b}{a}\hat{y}Y\right), \quad (4a)$$

$$\frac{d\hat{y}}{d\hat{t}} = \frac{c}{a}\hat{y} \left(1 - \frac{\hat{y}Y}{\eta_2} + \frac{d\eta_1}{c}\hat{x}\right), \quad (4b)$$

Then if $Y = \eta_2$, we are left with

$$\frac{d\hat{x}}{d\hat{t}} = \hat{x} \left(1 - \hat{x} - \frac{b\eta_2}{a}\hat{y}\right), \quad (5a)$$

$$\frac{d\hat{y}}{d\hat{t}} = \frac{c}{a}\hat{y} \left(1 - \hat{y} + \frac{d\eta_1}{c}\hat{x}\right), \quad (5b)$$

and so $\gamma_1 = b\eta_2/a$, $\beta = c/a$ and $\gamma_2 = d\eta_1/c$. Removing hats, we're left with

$$\frac{dx}{dt} = x(1 - x - \gamma_1 y), \quad (6a)$$

$$\frac{dy}{dt} = \beta y(1 - y + \gamma_2 x). \quad (6b)$$

All together:

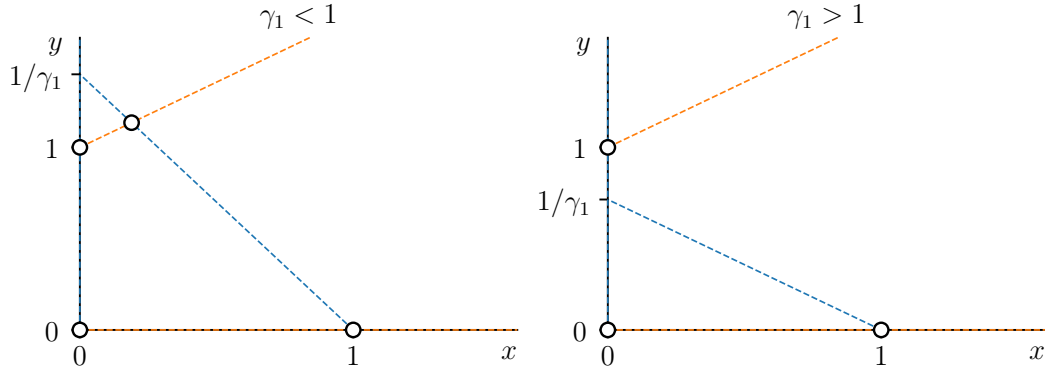
$$X = \eta_1, \quad Y = \eta_2, \quad T = \frac{1}{a}, \quad \beta = \frac{c}{a}, \quad \gamma_1 = \frac{b\eta_2}{a}, \quad \gamma_2 = \frac{d\eta_1}{c}. \quad (7)$$

3. The nullclines are

$$0 = x(1 - x - \gamma_1 y) \implies x = 0, \quad x = 1 - \gamma_1 y, \quad (8a)$$

$$0 = \beta y(1 - y + \gamma_2 x) \implies y = 0, \quad y = 1 + \gamma_2 x. \quad (8b)$$

Sketching these lines we see that whether the two diagonal lines cross to form a permissible equilibrium depends on $\gamma_1 \leq 1$:



4. Let's have a look at the four equilibria in turn.

Equilibrium 1, $(0, 0)$: The Jacobian is

$$J = \begin{pmatrix} 1 & 0 \\ 0 & \beta \end{pmatrix}, \quad (9)$$

and so the eigenvalues are $1, \beta > 0$ so this is always an unstable equilibrium (indeed an unstable node, but this question does not ask you to classify the equilibria more than stable/degenerate/unstable). So far:

	$(0, 0)$	$(0, 1)$	$(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$	$(1, 0)$
$\gamma_1 < 1$	unstable	?	?	?
$\gamma_1 = 1$	unstable	?	?	?
$\gamma_1 > 1$	unstable	?	?	?

Equilibrium 2, $(0, 1)$: The Jacobian is

$$J = \begin{pmatrix} 1 - \gamma_1 & 0 \\ \beta\gamma_2 & -\beta \end{pmatrix}, \quad (10)$$

and so the eigenvalues are $1 - \gamma_1$ and $-\beta < 0$. So this equilibrium is:

- $\gamma_1 < 1$: unstable (saddle point),
- $\gamma_1 = 1$: degenerate,
- $\gamma_1 > 1$: stable (stable node).

So far:

	(0, 0)	(0, 1)	$(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$	(1, 0)
$\gamma_1 < 1$	unstable	unstable	?	?
$\gamma_1 = 1$	unstable	degenerate	?	?
$\gamma_1 > 1$	unstable	stable	?	?

Equilibrium 4, (1, 0): The Jacobian is

$$J = \begin{pmatrix} -1 & -\gamma_1 \\ 0 & \beta(1 + \gamma_2) \end{pmatrix}, \quad (11)$$

and so the eigenvalues are -1 and $\beta(1 + \gamma_2) > 0$. So this equilibrium is always unstable (a saddle point). So far:

	(0, 0)	(0, 1)	$(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$	(1, 0)
$\gamma_1 < 1$	unstable	unstable	?	unstable
$\gamma_1 = 1$	unstable	degenerate	?	unstable
$\gamma_1 > 1$	unstable	stable	?	unstable

Equilibrium 3, $(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$: Time to face the music. The question tells us that the eigenvalue equation is

$$c_2\lambda^2 + c_1\lambda + c_0 = 0, \quad (12)$$

where

$$c_2 = 1 + \gamma_1\gamma_2, \quad (13)$$

$$c_1 = 1 - \gamma_1 + \beta[1 + \gamma_2], \quad (14)$$

$$c_0 = \beta(1 - \gamma_1)(1 + \gamma_2), \quad (15)$$

First, notice that when $\gamma_1 = 1$, this equilibrium is equal to $(0, 1)$, which we already know is degenerate in this case.

We can also notice that if $\gamma_1 > 1$, then this equilibrium is not permissible, but we can still do a linear stability analysis on it.

Let's look at the signs of $c_{2,1,0}$ for the two unsolved cases of γ_1 , remembering that the question tells us that $\gamma_1 > 0$:

- $(0 <) \gamma_1 < 1$:

$$c_2 > 0, \quad c_1 > 0, \quad c_0 > 0.$$

By Descartes' rule of signs, $\text{Re}(\lambda_1), \text{Re}(\lambda_2) < 0$, so this equilibrium is stable.

- $\gamma_1 > 1$:

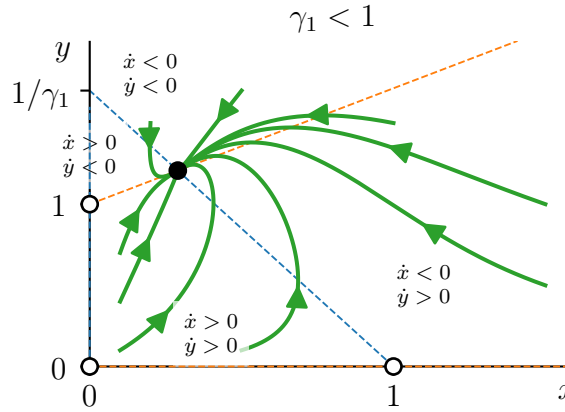
$$c_2 > 0, \quad c_1 \dots?, \quad c_0 < 0.$$

We don't need to know the sign of c_1 to see that there can only be one change of sign, so $\text{Re}(\lambda_1) < 0 < \text{Re}(\lambda_2)$ and so this is a saddle point, and hence unstable.

In conclusion:

	$(0, 0)$	$(0, 1)$	$(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$	$(1, 0)$
$\gamma_1 < 1$	unstable	unstable	stable	unstable
$\gamma_1 = 1$	unstable	degenerate	equal to $(0, 1)$	unstable
$\gamma_1 > 1$	unstable	stable	unstable	unstable

5. The phase portrait should look like:



showing the stability of the $(\frac{1-\gamma_1}{1+\gamma_1\gamma_2}, \frac{1+\gamma_2}{1+\gamma_1\gamma_2})$ equilibrium.

6. As you noted in question 1, this model is not a pure predator–prey one. The fact the the ‘predator’ has logistic growth in the absence of prey indicates it has an alternate food source to x as it will reach a non-zero equilibrium even if x is zero. The interaction terms could instead reflect a competitive system where y carries a fatal disease for x , so interactions make the alternative food source more accessible for y . So actually, x is less well able to survive rather than being directly eaten!

- Unstable $(1, 0)$ equilibrium: this says the weaker species cannot win out or outcompete. I think you are relatively safe to say this is a good thing. I am sure one could find examples in nature where the weaker competitor can win out, but it would be a very extreme system.
- No stable $(0, 0)$ equilibrium: Many students either class this as invalid or valid. In reality I think it’s ambiguous. Instability implies a small seeding population(s) will automatically grow to a finite (noticeable) value. In this model this behaviour is dictated by the logistic growth which we know on its own always forces a population to a finite level (the carrying capacity). For a highly infectious disease like Covid-19 this is probably a good property. But there will be some physical situations where some critical mass is required for a population to not die out quickly. Like we saw in the Allee model in 1D, this is known to be true of plankton populations; they survive far better in large populations due to their mutual cooperation and fail if it is lacking.
- The $(0, 1)$ and coexistence equilibrium switch stability when γ_1 goes from < 1 to > 1 : This is the best aspect of in the model. We see that γ_1 represents the relative balance between growth of the weaker species x (from the logistic

model) and a negative interaction. When $\gamma_1 > 1$ the stronger competitor/-predator is too dominant and wins out (again this is not a problem, logistic growth says it has a separate food source); $\gamma_1 < 1$ implies a balance.

- One stable equilibrium: this is again a little ambiguous; we saw the notion of multistability in the outbreak/spruce budworm model in class. This implies the end state depends on where you start, i.e. for a disease if the initial seeding is not too big it may settle at a low level, but if it gets too big the disease can break out to a larger value. This is seen as a good model property in most of epidemiology, but our model here cannot have that. In the model at hand the end state is independent of the initial conditions. It's not a terrible property in and of itself, but it means the model is limited as to what physical situations it can describe.

Overall we are left with the conclusion this model has some good points, with a parametric dependence on who wins, but it is limited, with no multistability or mutual extinction. Of course it has no spatial behaviour as well which is a generic complaint I would always accept about any purely temporal system.